Joint ICTP-IAEA Workshop on Monte Carlo Radiation Transport and Associated Data Needs for Medical Applications
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Lecture 27

EGSnrc transport in a magnetic field

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Acknowledgment

This talk is based almost entirely on the work of Victor Malkov, a doctoral student in my group at Carleton until 2017. Now at Mayo Clinic, MN

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EGSnrc transport in magnetic fields

- Bielajew did this for EGS4 in the 1980s
  - EGS4 and the magnetic field transport required short steps
- these have been ported to EGSnrc by Ernesto
  $\Rightarrow$but the macros are not optimized

- EGSnrc takes longer steps and uses a single scattering mode to cross boundaries

magnetic transport

- standard starting point for change in direction due to magnetic field

$$\Delta\vec{u}_B(t) = \frac{qc^2}{v_o E}\int_0^t dt'\vec{v}(t') \times \vec{B}$$

- define

$$\boxed{\delta_u = |\Delta\vec{u}|}$$

- for little energy loss, constant magnetic field B and a small change in direction, a 1st order approximation is

$$\Delta\vec{u}_{B1} = \frac{qtc^2}{v_o E}\left(\vec{v}(0) \times \vec{B}\right)$$

1st order approximation: 1-PI

- converting time variable to pathlength, s

$$t = \frac{s}{v_o}\left(\frac{1}{2} + \frac{v_o}{2v_f}\right) = \frac{s}{v_o}\eta \Rightarrow \frac{s}{v_o}$$

gives

$$\Delta\vec{u}_{B1} = \frac{qtc^2}{v_o E}\left(\vec{v}(0) \times \vec{B}\right) = \frac{qs\eta}{\beta_o^2 E}\left(\vec{v}(0) \times \vec{B}\right)$$

where $\beta^2 = v^2/c^2$

This equation is what is often used.

3-point integration: 3-PI

- accuracy can be improved by doing a 3 point integration of general eqn and making same substitutions for t

$$\Delta\vec{u}_{B3} = \frac{qs\eta}{\beta_o^2 E}\frac{1}{6}\left(\vec{v}(0)\times\vec{B} + 4\vec{v}\left(\frac{t}{2}\right)\times\vec{B} + \vec{v}(t)\times\vec{B}\right)$$

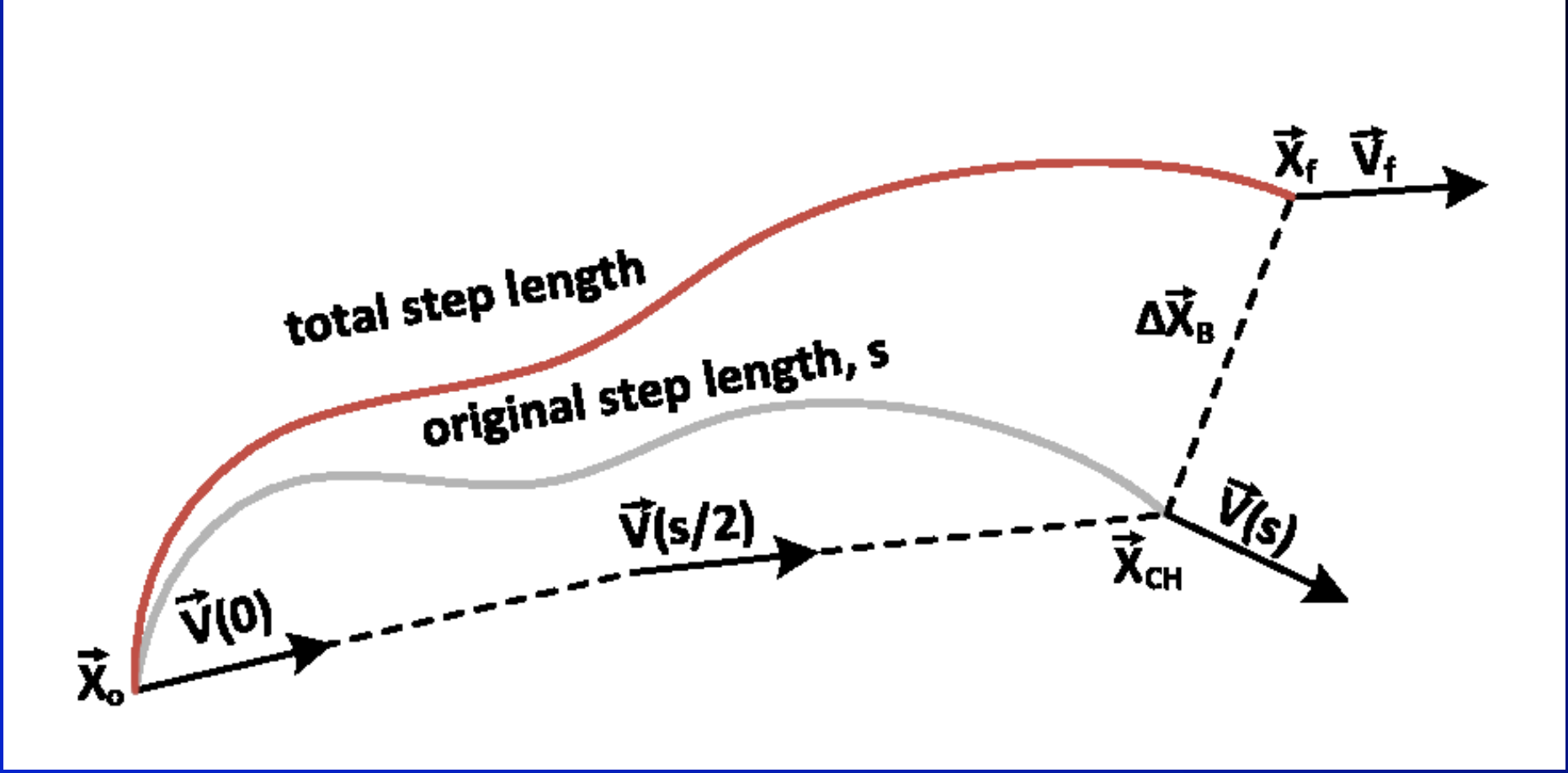
- change in position is given by

$$\Delta\vec{x}_B = \frac{s}{2}\left(\frac{1}{2} + \frac{v_o}{2v_f}\right)\Delta\vec{u}_B = \frac{s\eta}{2}\Delta\vec{u}_B$$

- max step size from δ_u restriction (none for B=0)

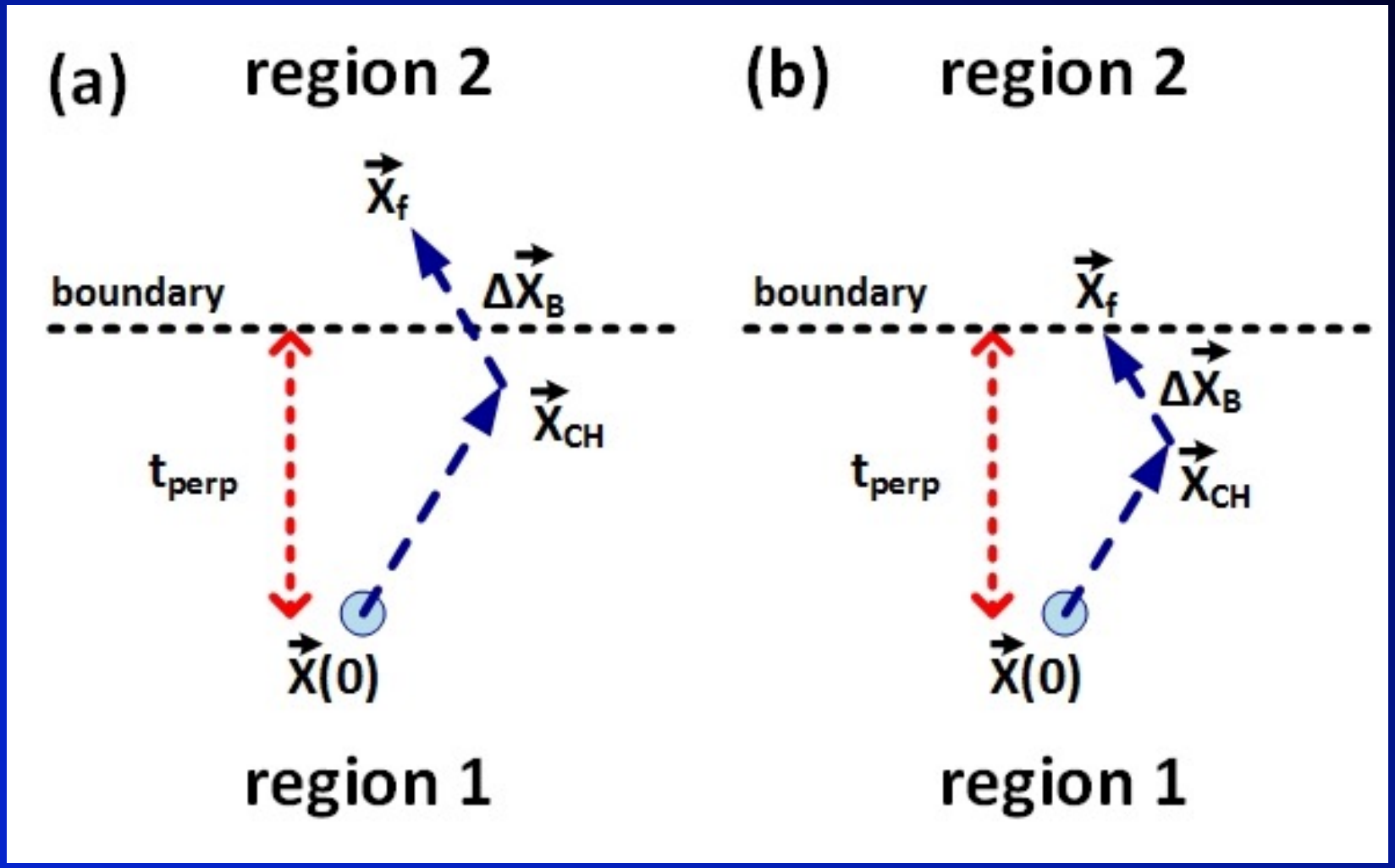
$$s(B,\delta_u,E_o) = \frac{\delta_u\beta^2 E_o}{|q\vec{v}(0)\times\vec{B}|}$$

effect of magnetic field



EGSnrc transport is more complex, but this illustrates additional magnetic effects

t_{perp} no longer prevents crossing boundary



$$t_{perp} = t_{CH,max} + |\Delta\vec{x}_B|$$

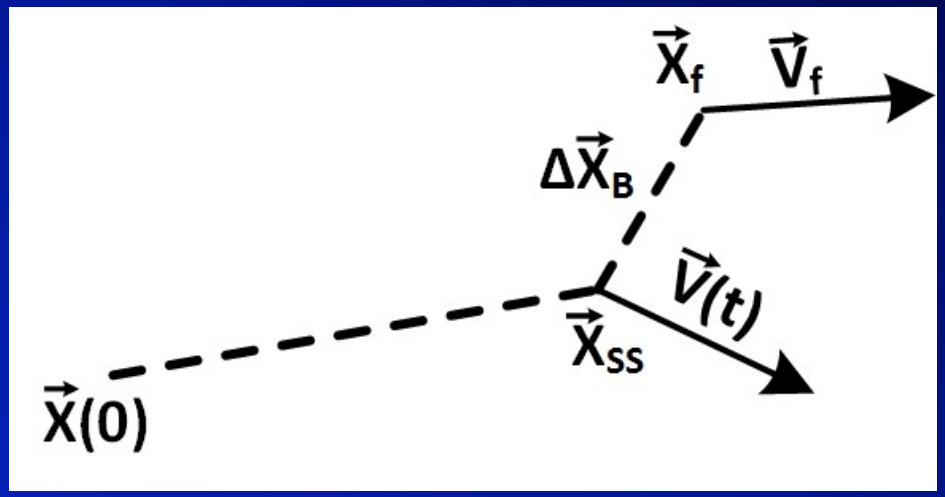
single scattering mode: SS

- if a particle < skin depth from any boundary
  - code goes into single scattering mode
    - default EGSnrc transports in straight lines to interaction and then scatters, or to boundary where it stops, resamples and continues to next interaction.

- with a mag field there are several issues or approaches

single scatter transport options

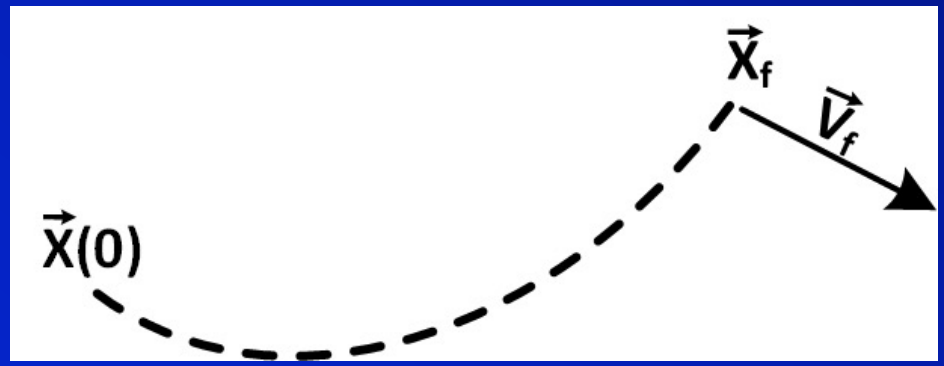
SS-1st



1st O calc as in CH step.

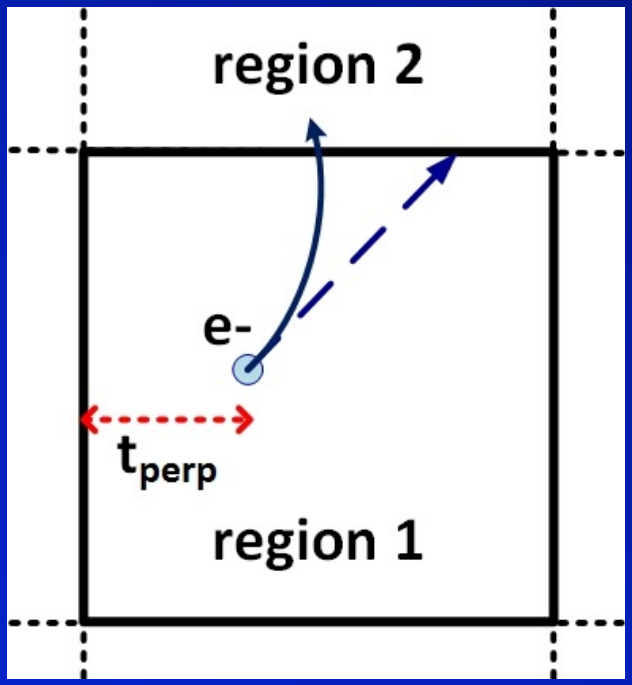
But leads to significant breakdowns

SS-analytic

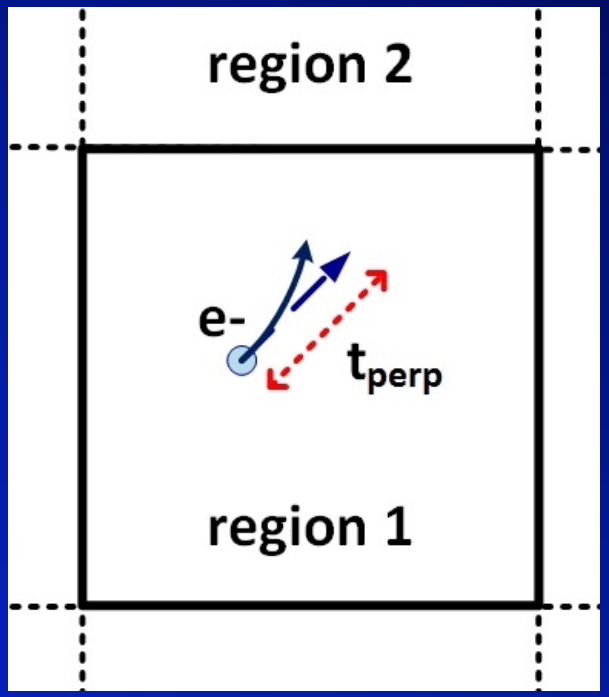


analytic sol'n for transport in vacuum. Always used

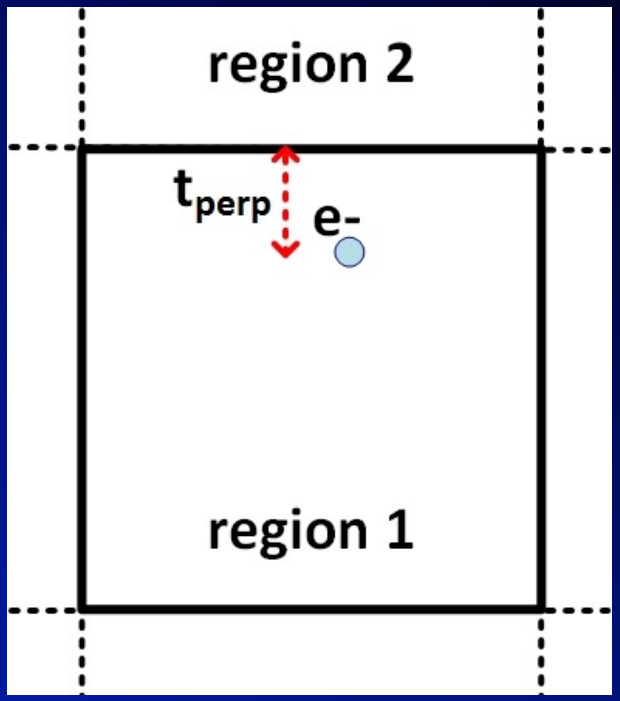
boundary crossing algorithm: BCA_{perp}



SS mode: step to boundary could go to wrong region



so shorten it to t_{perp}

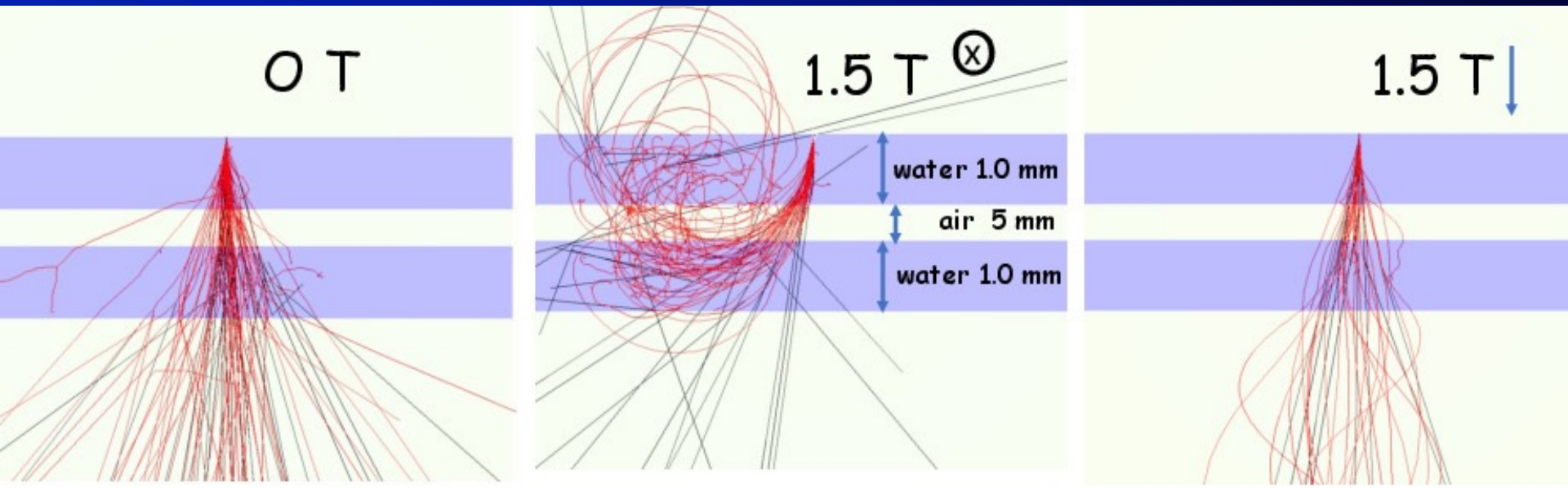


below some limit (default 10 nm) just ignore B

guaranteed to work(slower)

B field effects on electrons

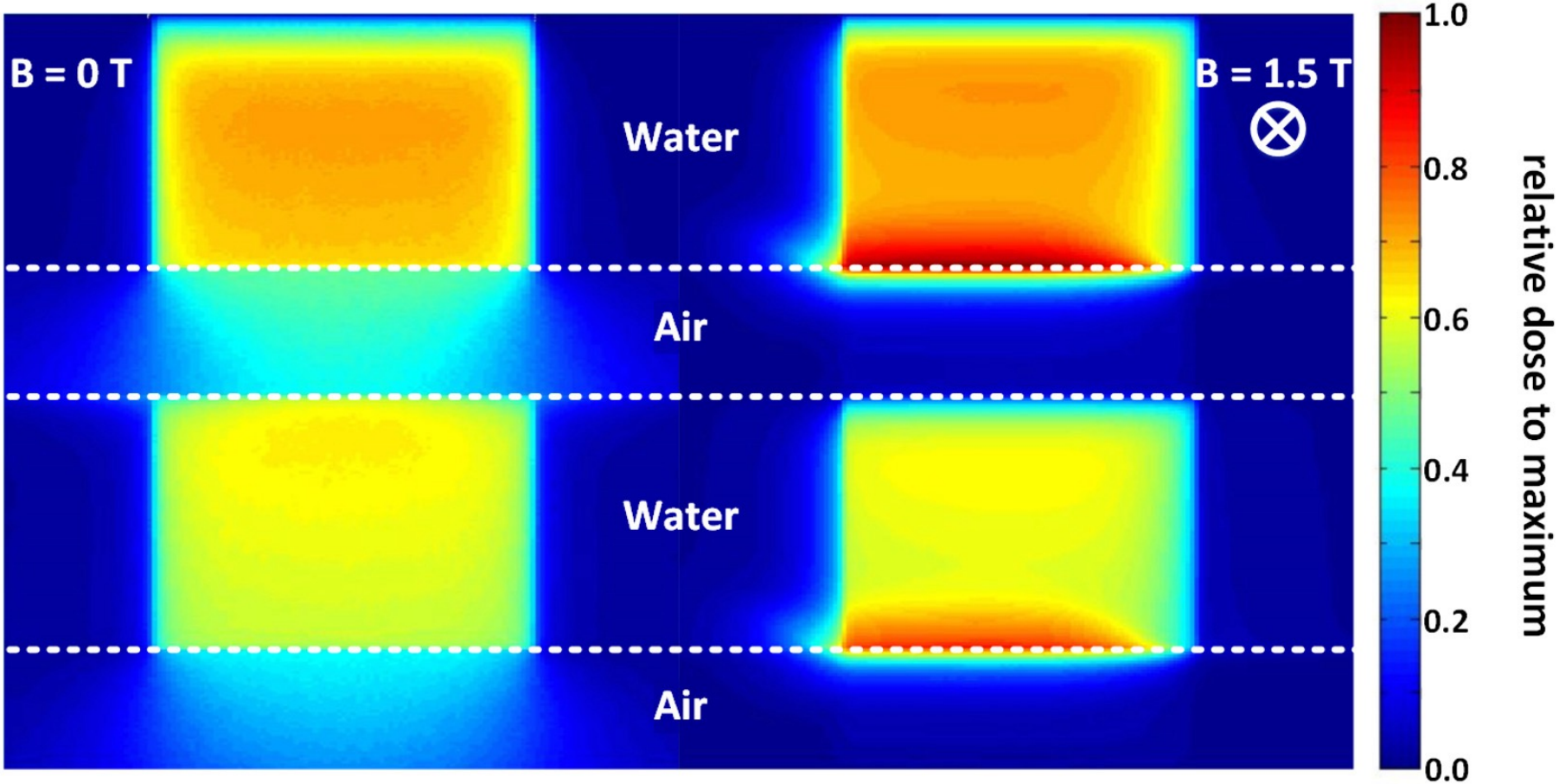
pencil beams of 10 MeV e-



electron return effect

Magnetic field effects: Electron Return Effect (ERE)

photons on water phantom with air gap

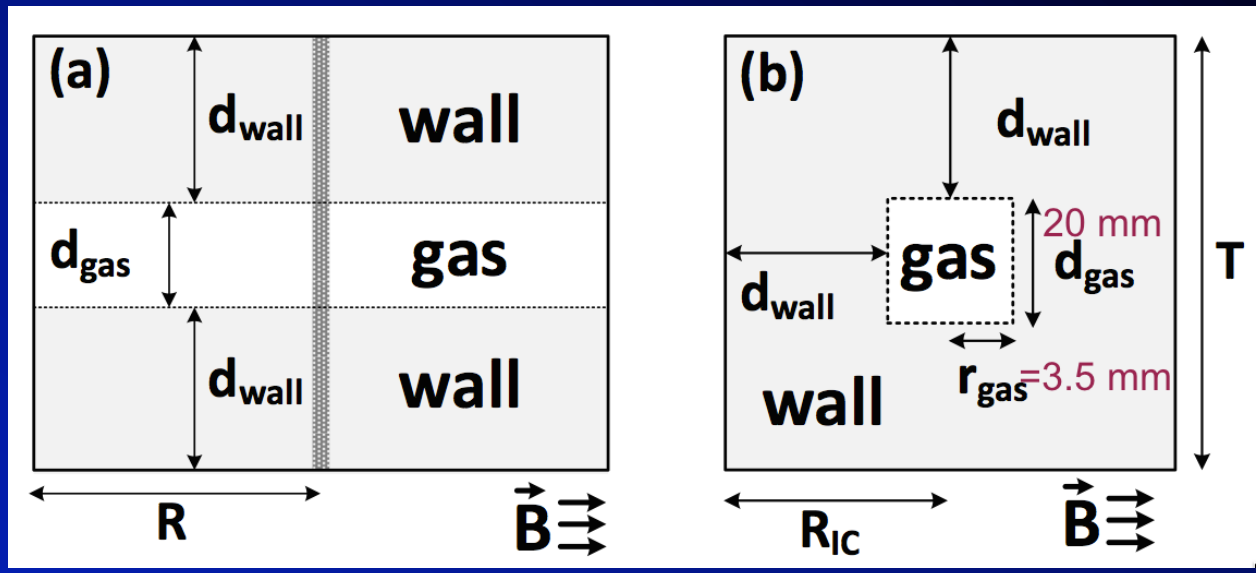


Fano test with magnetic field

- the Fano test of a Monte Carlo code is a rigorous test of its ability to simulate ion chamber response accurately
  - ion chamber response is the most difficult simulation to do
- Bouchard et al (PMB 60(2015)6639) have proposed a version valid in a magnetic field
  - with a uniform B, the electron source needs to be isotropic and uniform per unit mass
  - geometry is 2 semi-infinite slabs separated by a gas of same material or ion chamber volume

Fano test (continued)

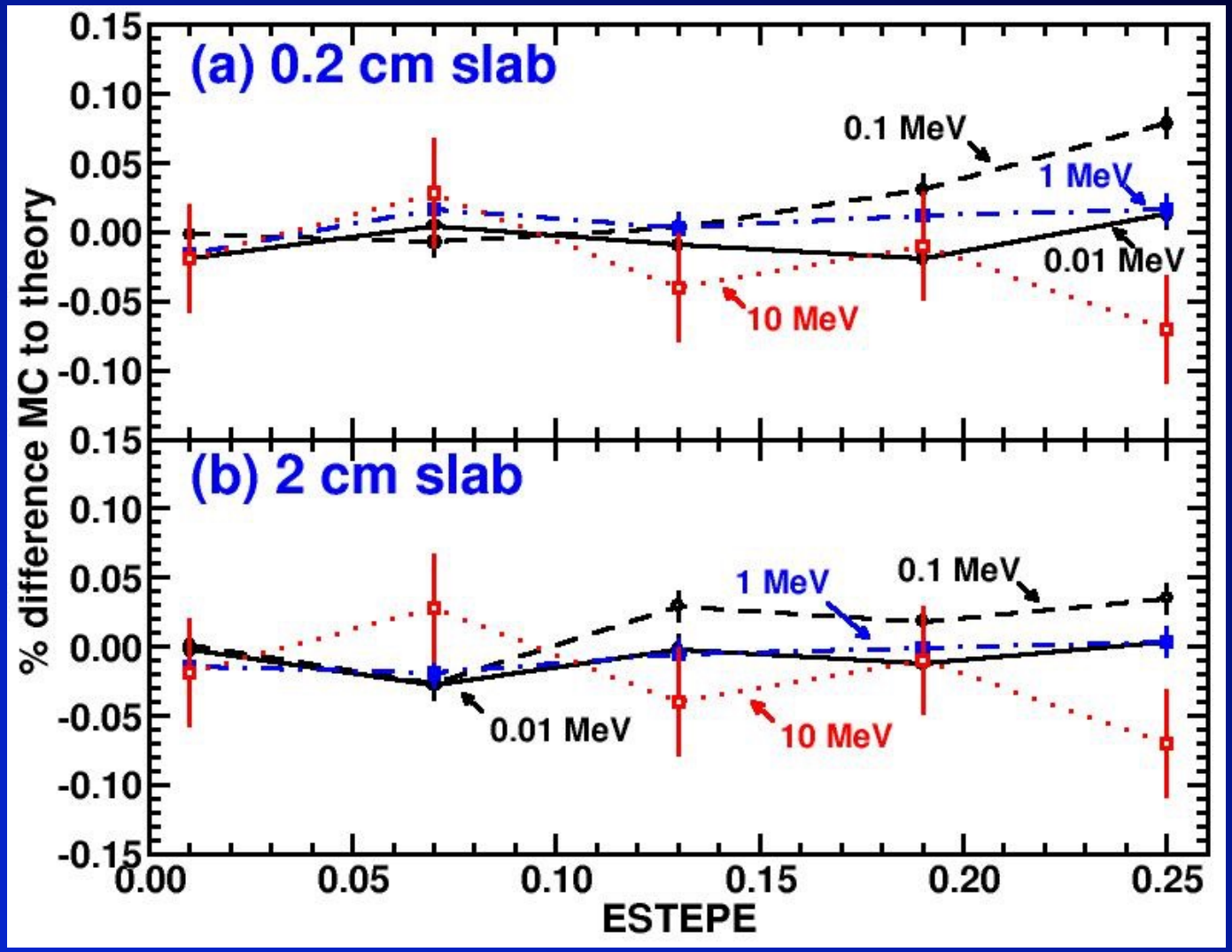
- force photons to deposit energy on spot
- reciprocity theorem
  - isotropic line source equivalent to isotropic sources everywhere



$$\mathrm{MC(dose\ in\ gas)} \stackrel{?}{=} I E_o$$

I is number per unit mass of initial electrons of energy E_o

EGSnrc Fano test with no B field



isotropic electron sources

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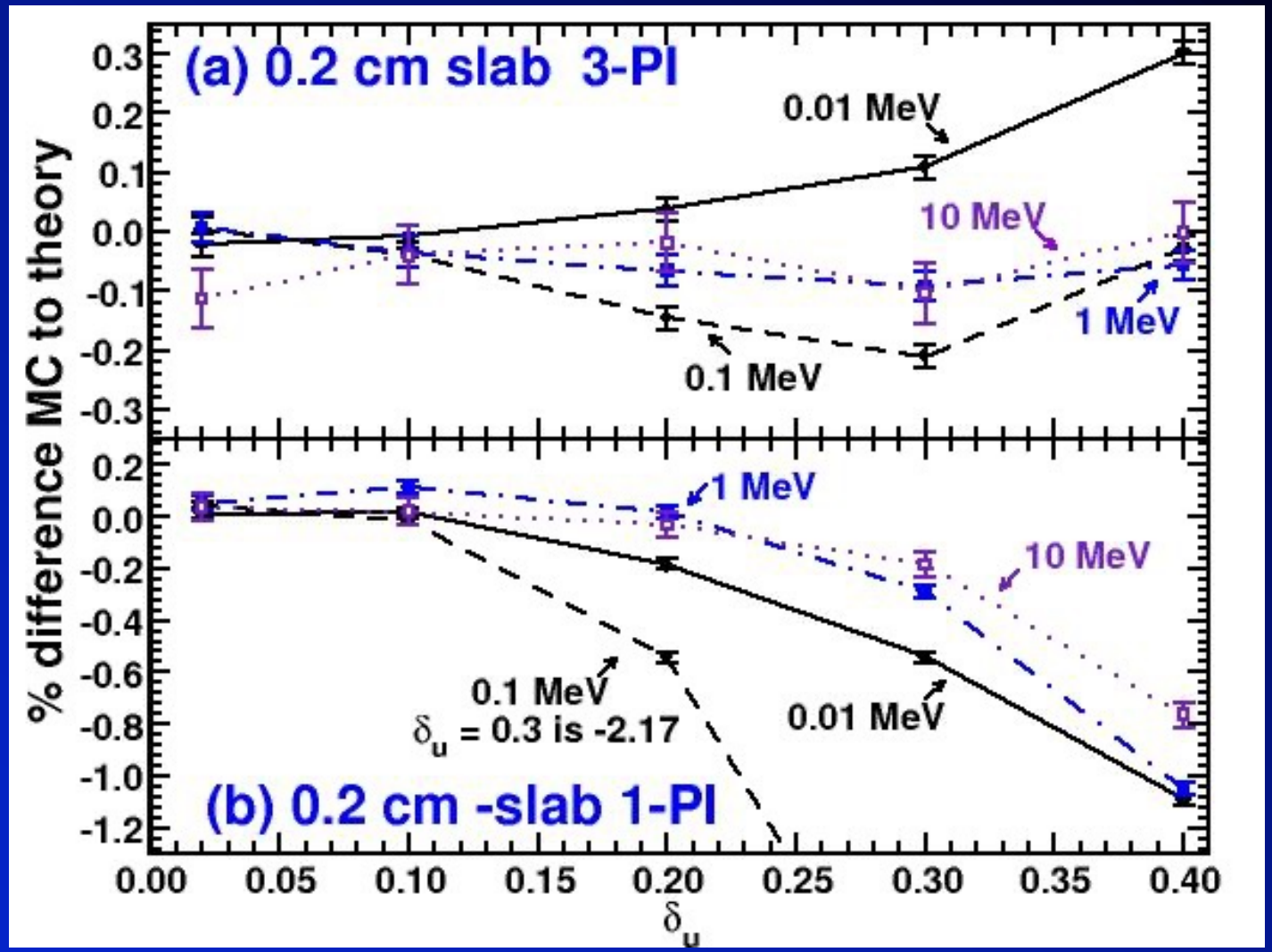
EGSnrc Fano test 1.5 T: 2 mm gap

B || gap.

harder test than B⊥gap

note different scales

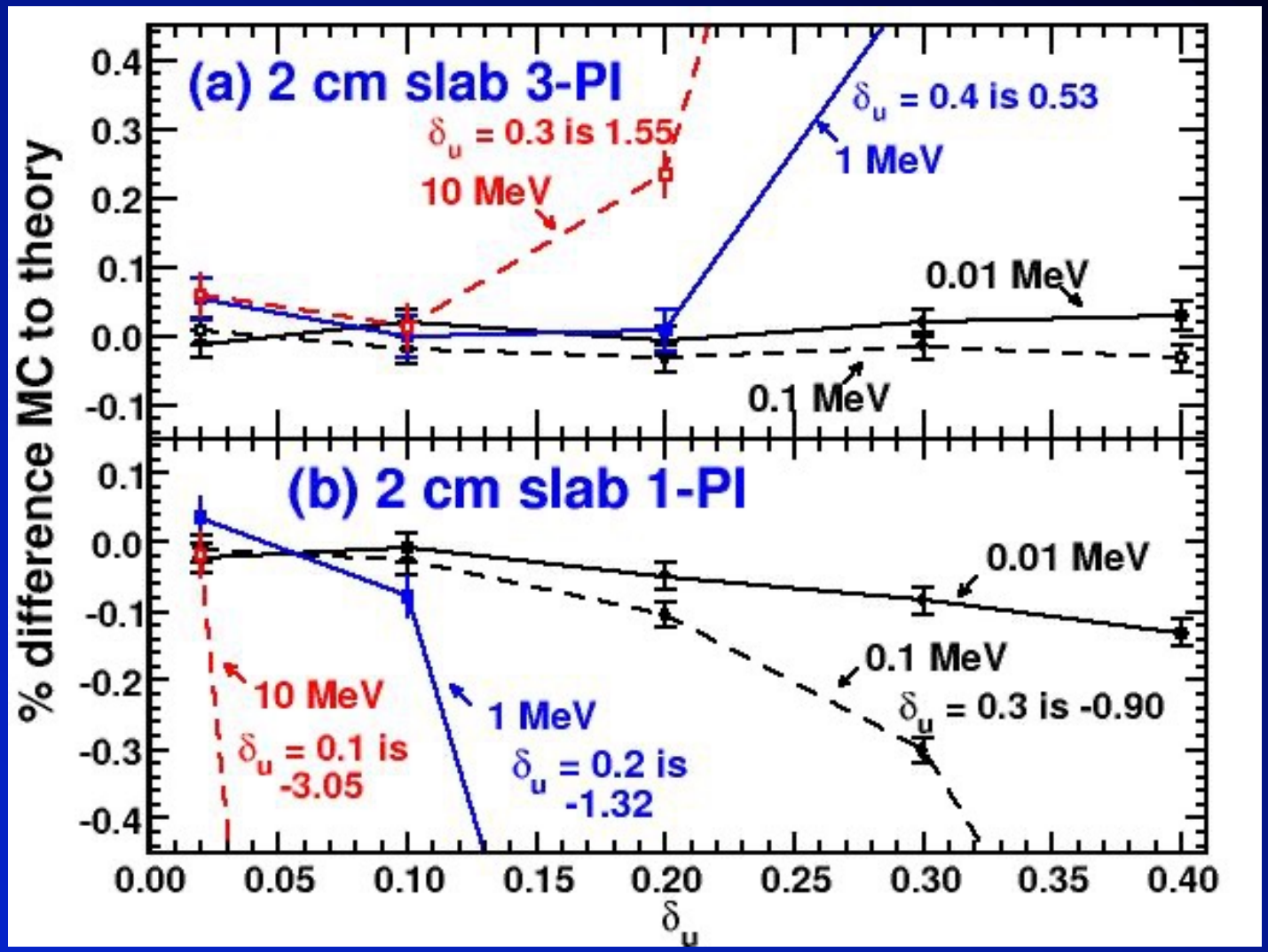


EGSnrc Fano test 1.5 T: 2 cm gap

B || gap.

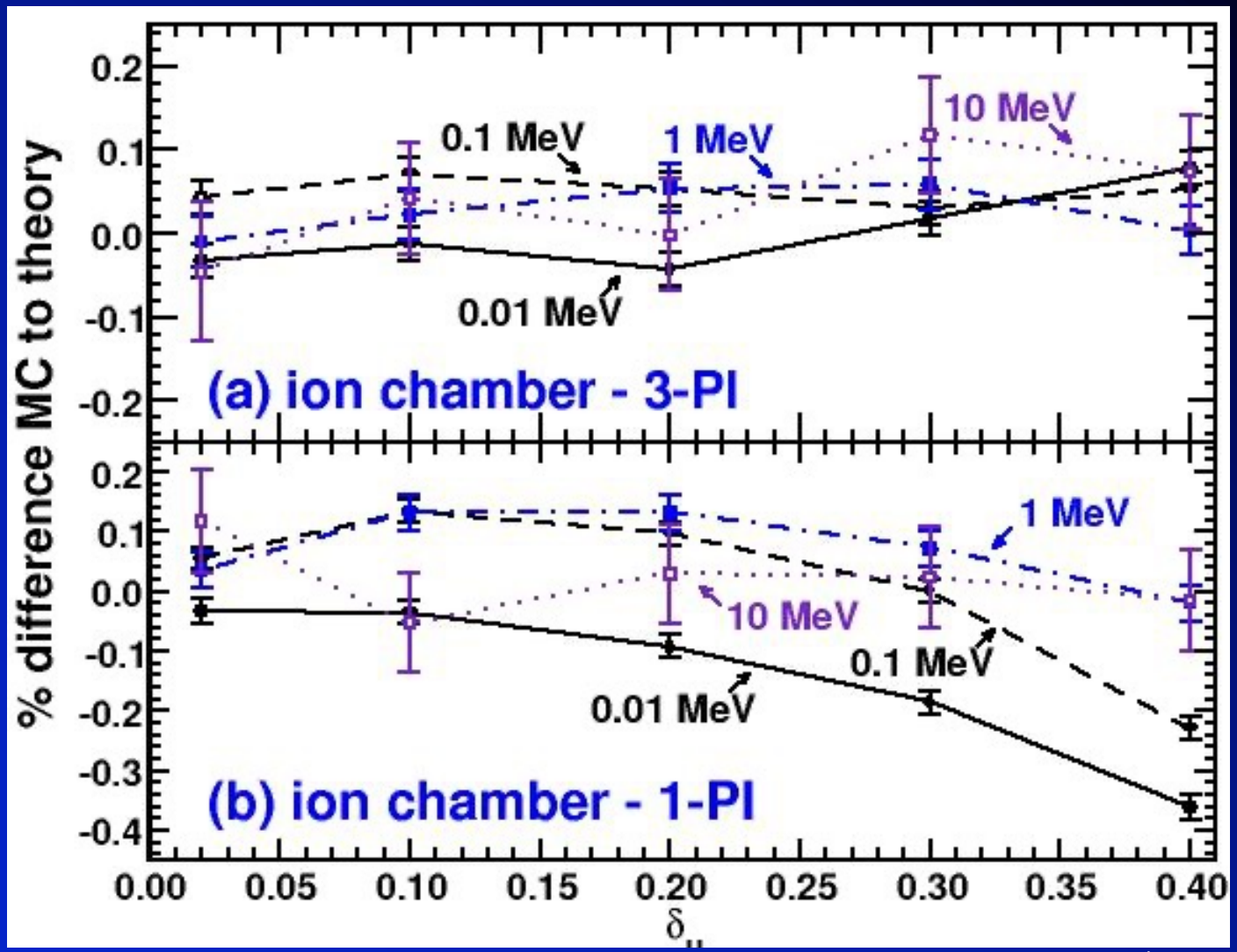
harder test than B⊥gap

scales same



EGSnrc Fano test 1.5 T: ion chamber

The 1-PI passes here but failed 2 cm slab version



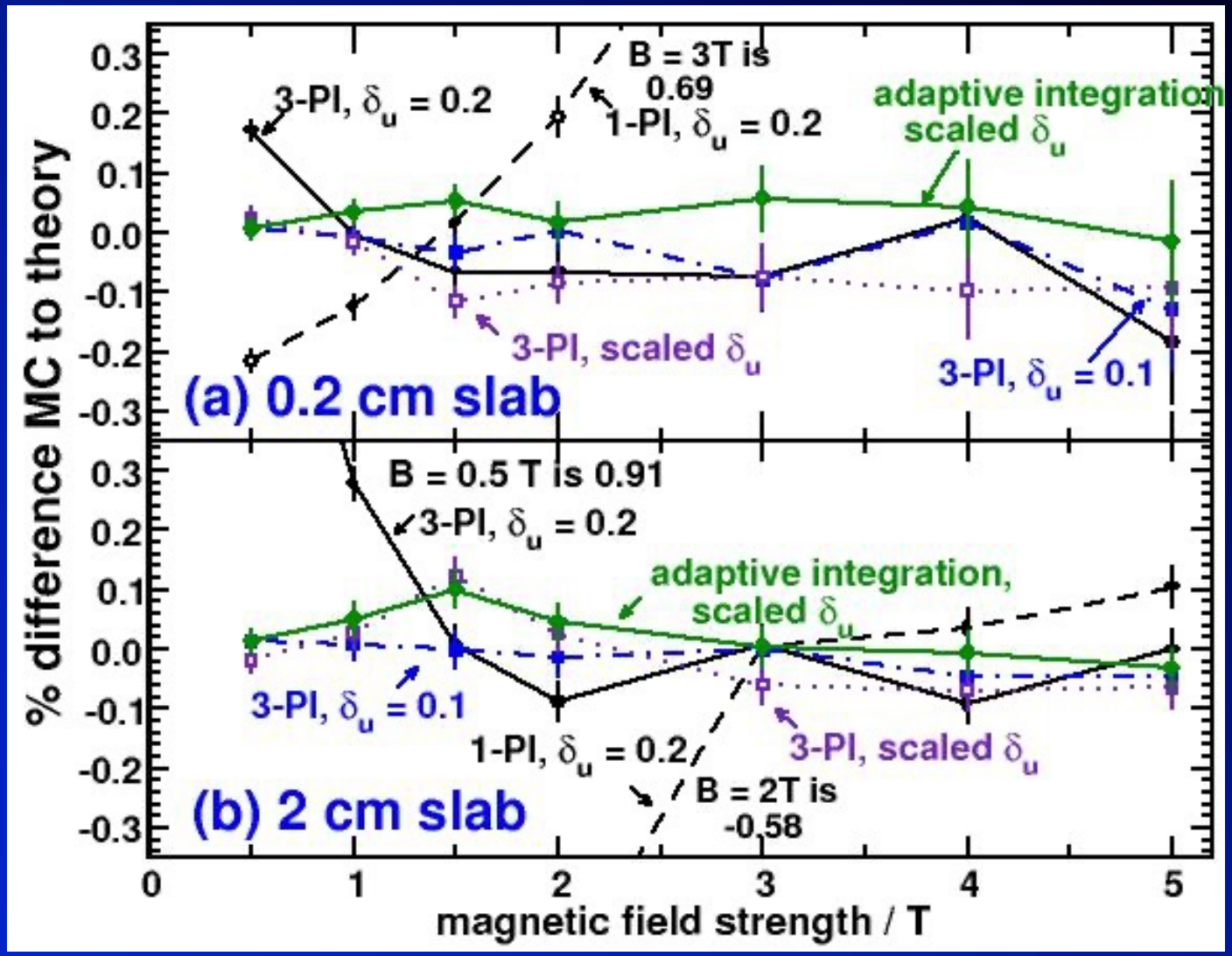
Fano slab gap test vs B: 1 MeV e-

adaptive integration

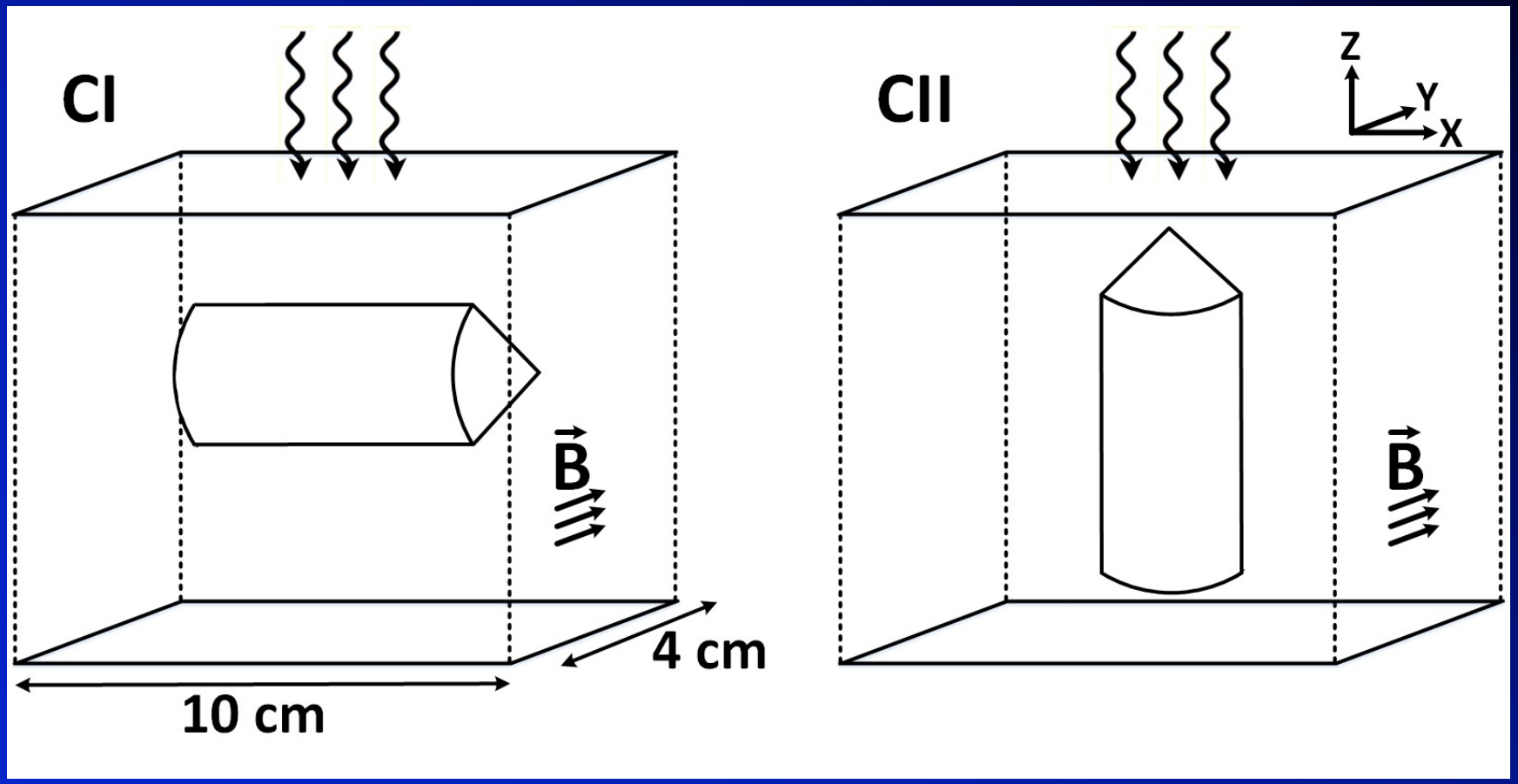
-uses 1-PI for short steps

scaled δ_u

varies with B relative to B_{ref} value



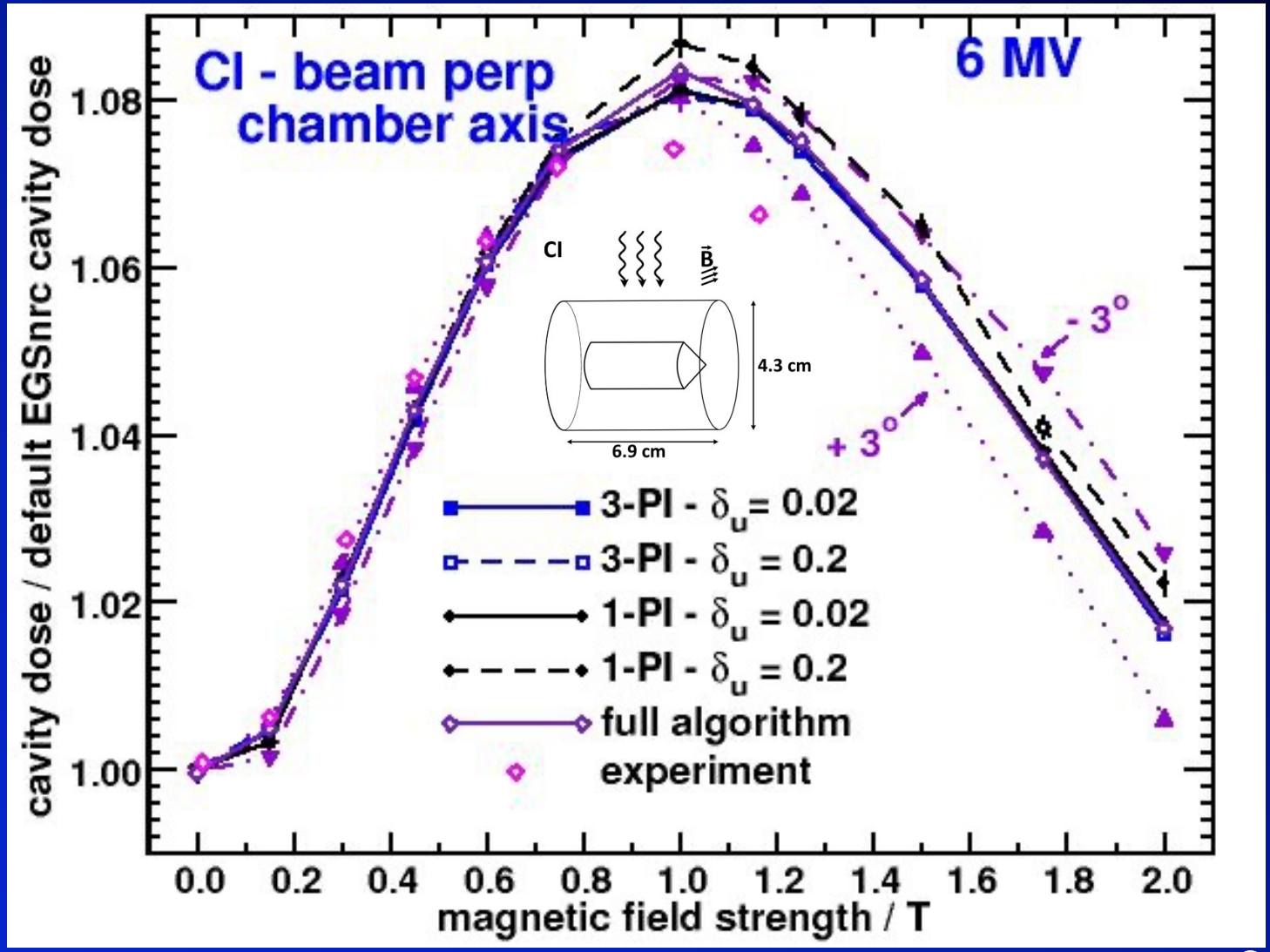
ion chamber simulations



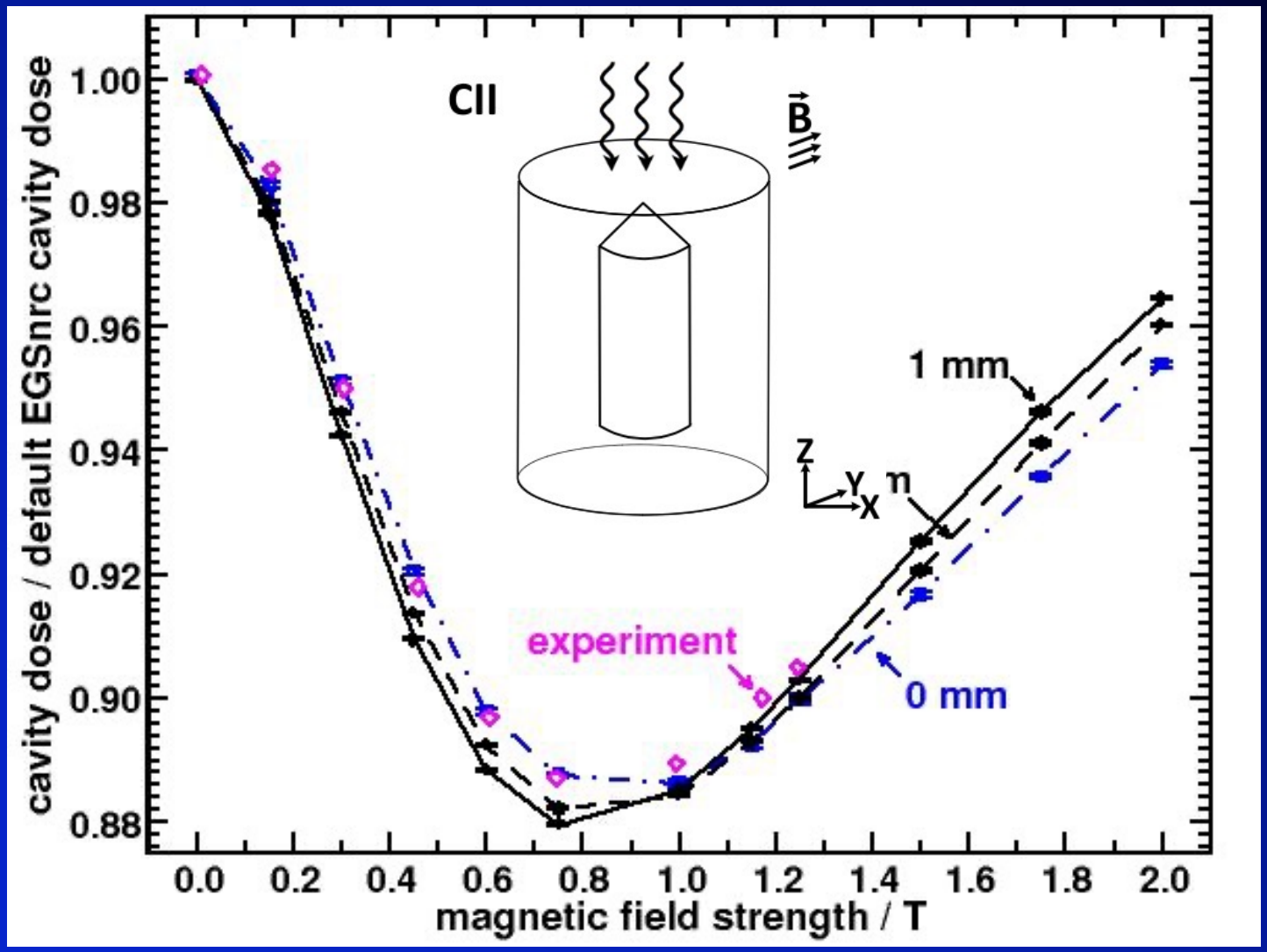
after Meijsing et al PMB 54(2009) 2993

NE2571 models were detailed.

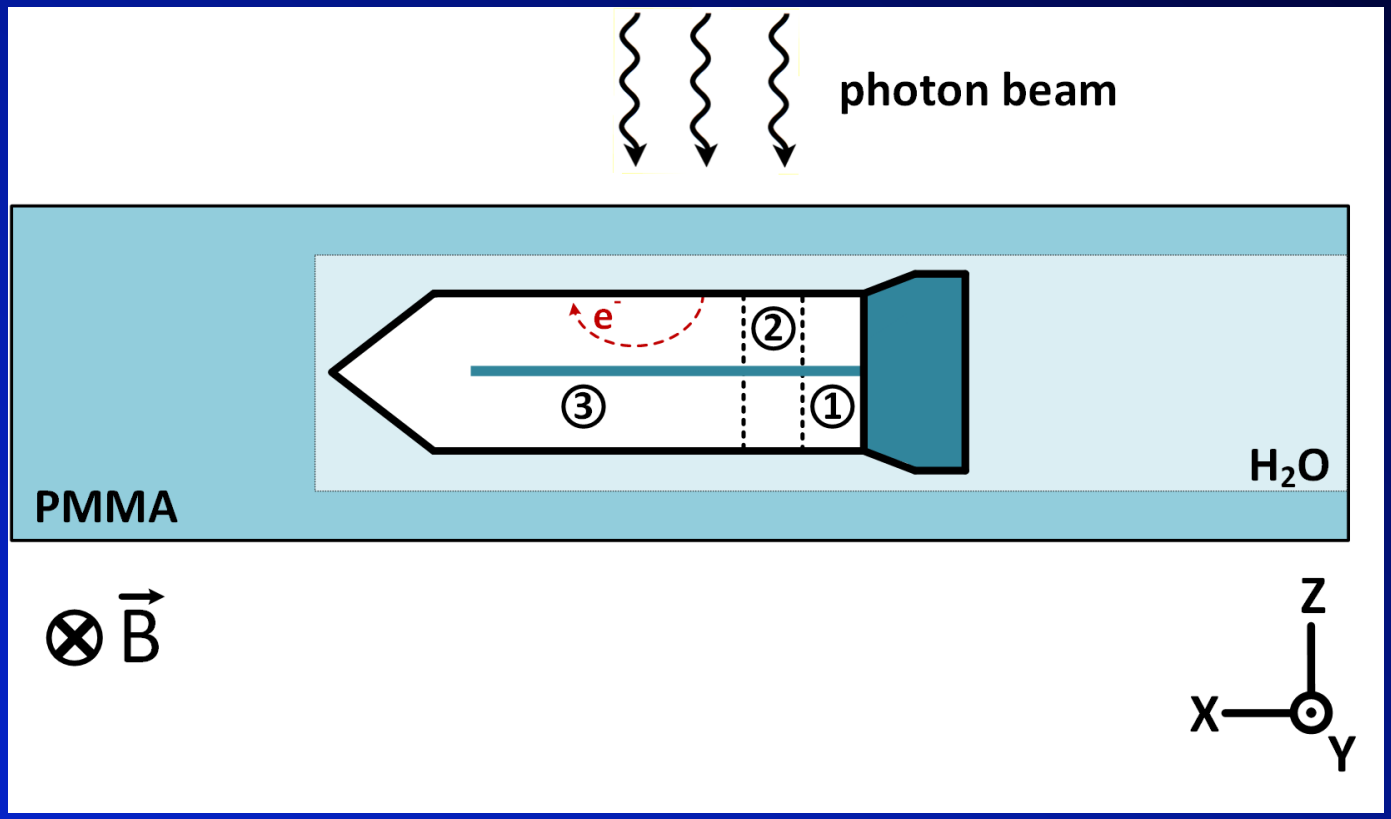
C I geometry: beam & B ⊥ chamber axis



C I I geometry: beam || axis & both ⊥ B
air gap effects



Complications in a B-field



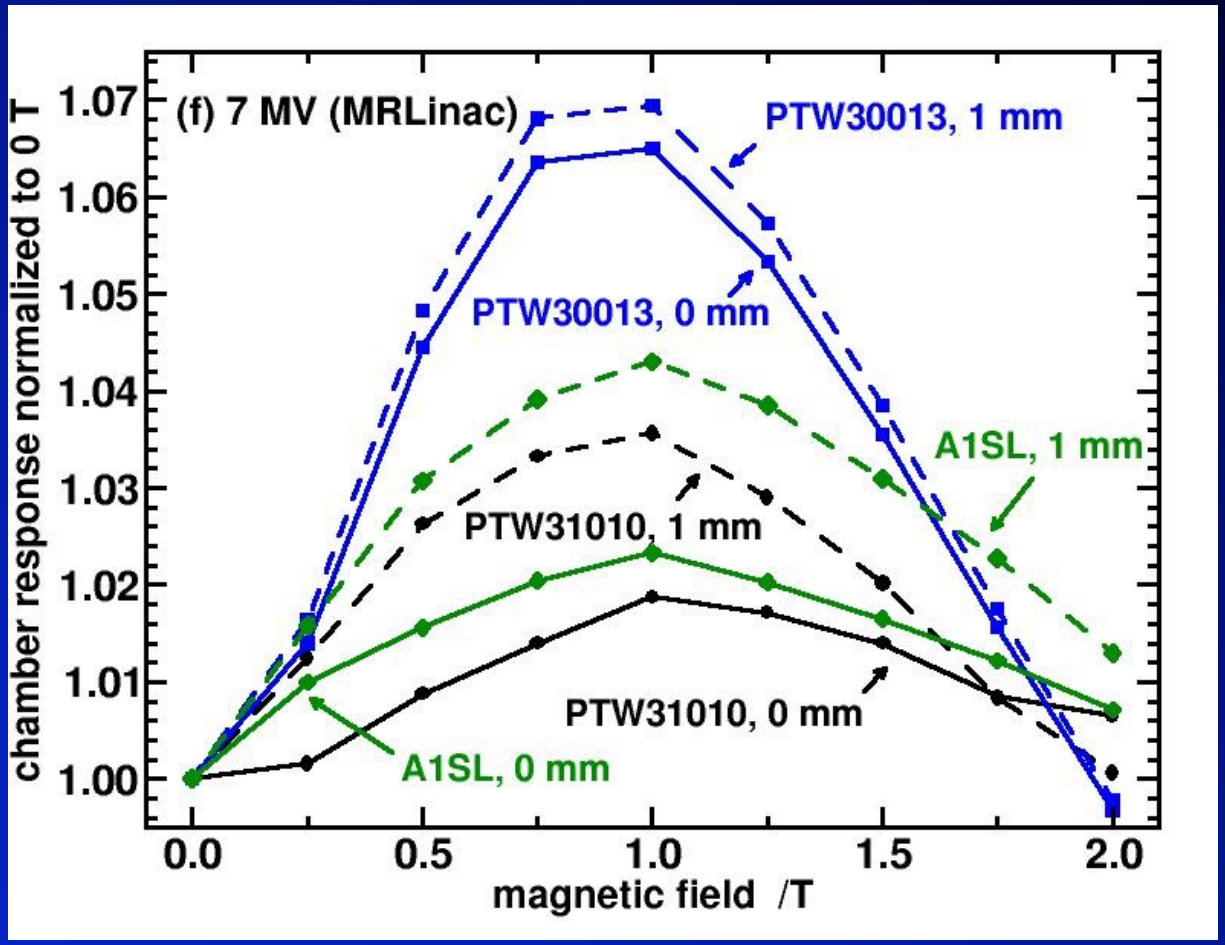
Ion chambers have dead regions near stem due to E-field distortions

Volume uncertain.

Malkov Med Phys 44 (2017) 4854

Complications in a B-field

beam & B-field perpendicular to chamber



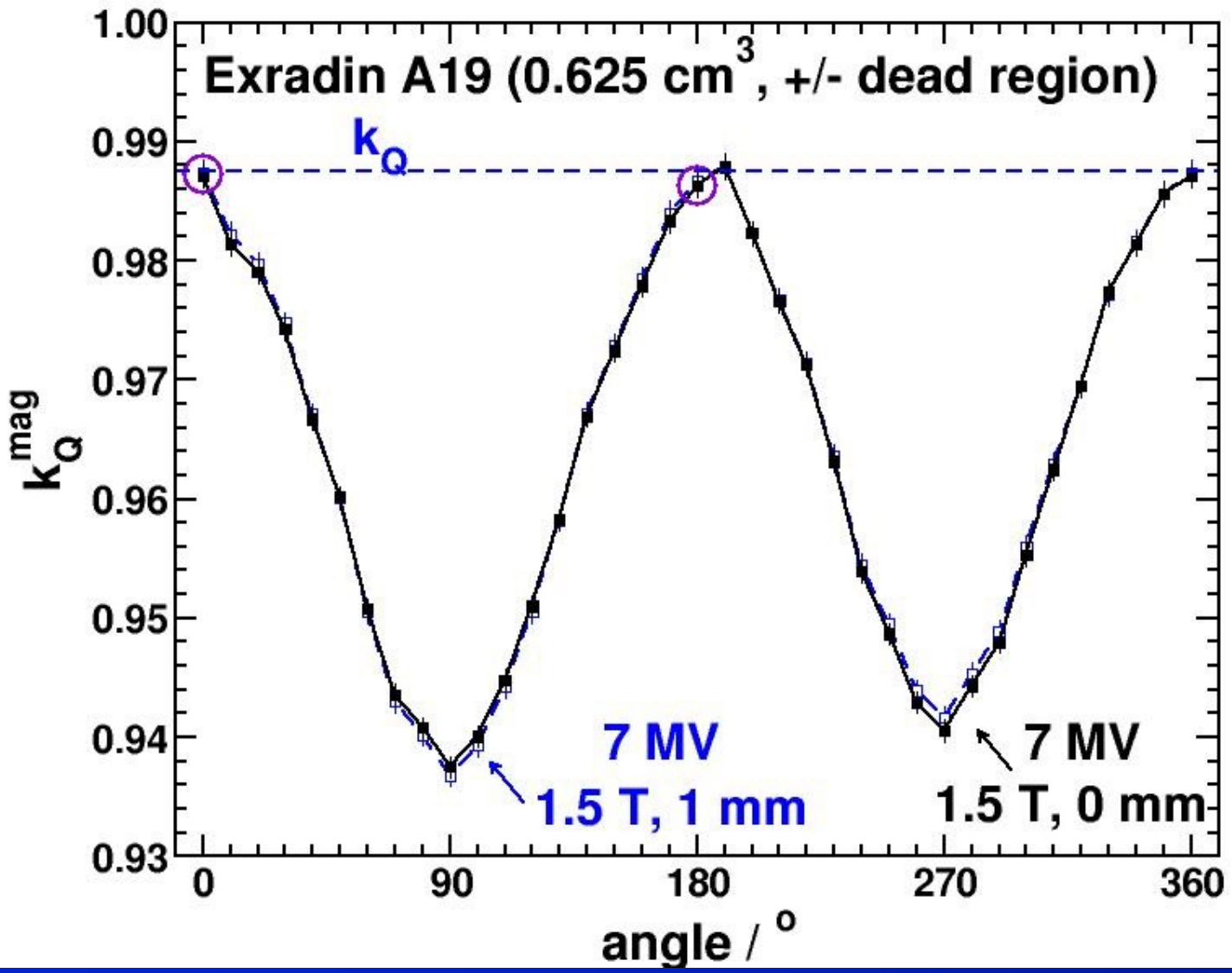
Worst case effects from uncertain sensitive region

$k_Q^{mag} = k_Q \, k_{B,Q}$

Photons perpendicular to mag field.

Angle of chamber axis wrt to mag field

Dead air thickness shown

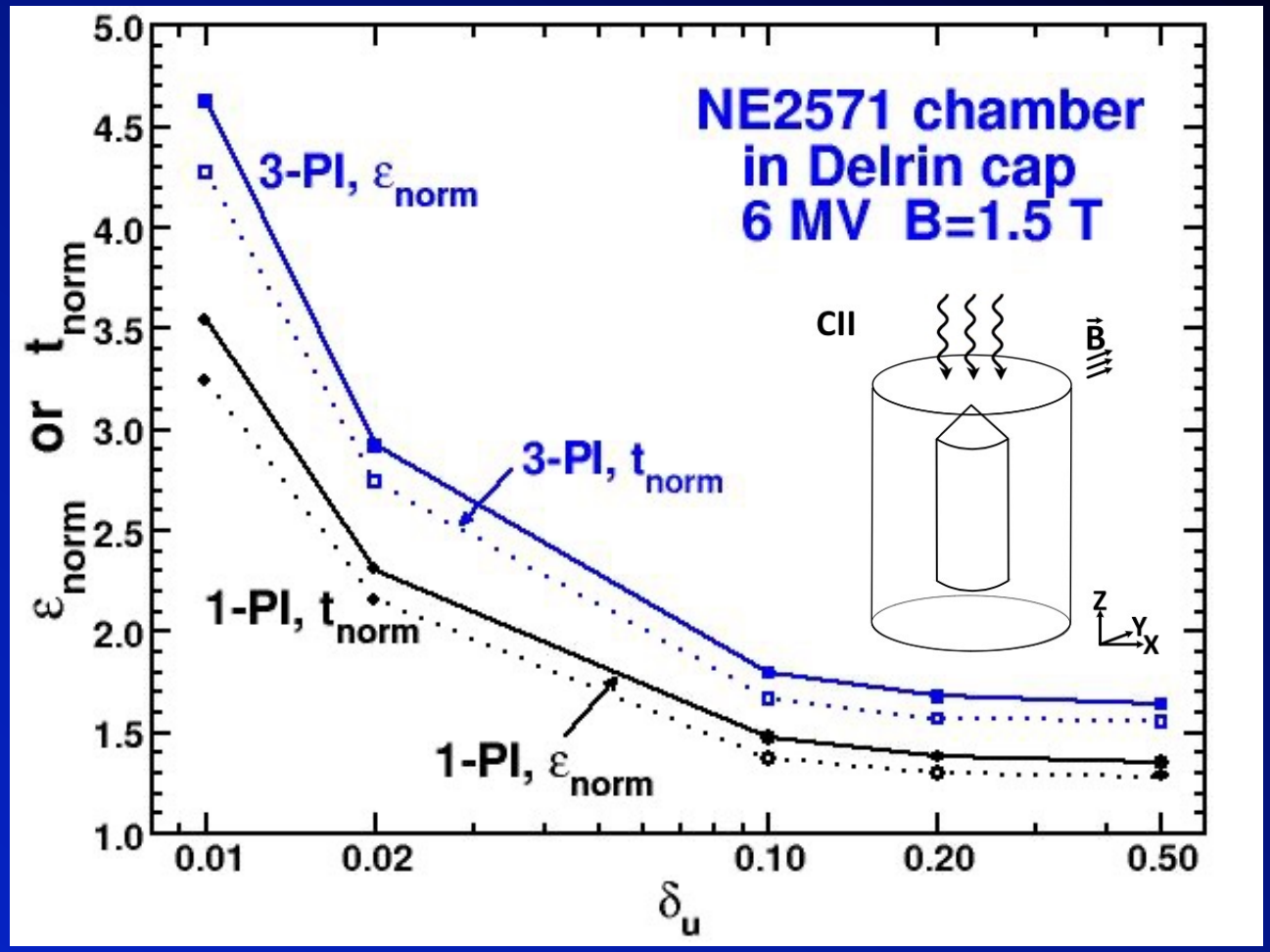


calculation efficiency: ion chamber

CII case, beam || chamber axis

$$t_{\mathrm{norm}} = \frac{t_B}{t_0}$$

$$\epsilon_{\mathrm{norm}} = \frac{\epsilon_0}{\epsilon_B} = t_{\mathrm{norm}} \frac{s_B^2}{s_0^2}$$



smaller t_{norm} better: ε_{norm} is inverted,

higher ε better => lower ε_{norm} is better

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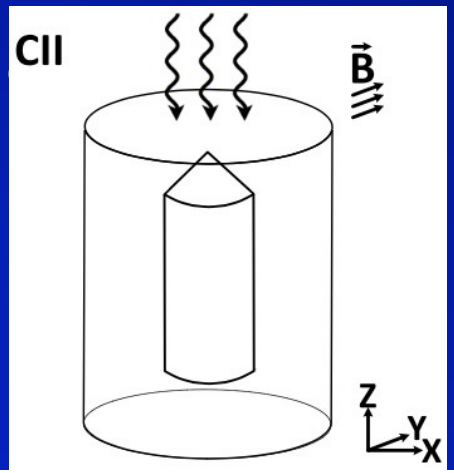
calculation efficiency: ion chamber



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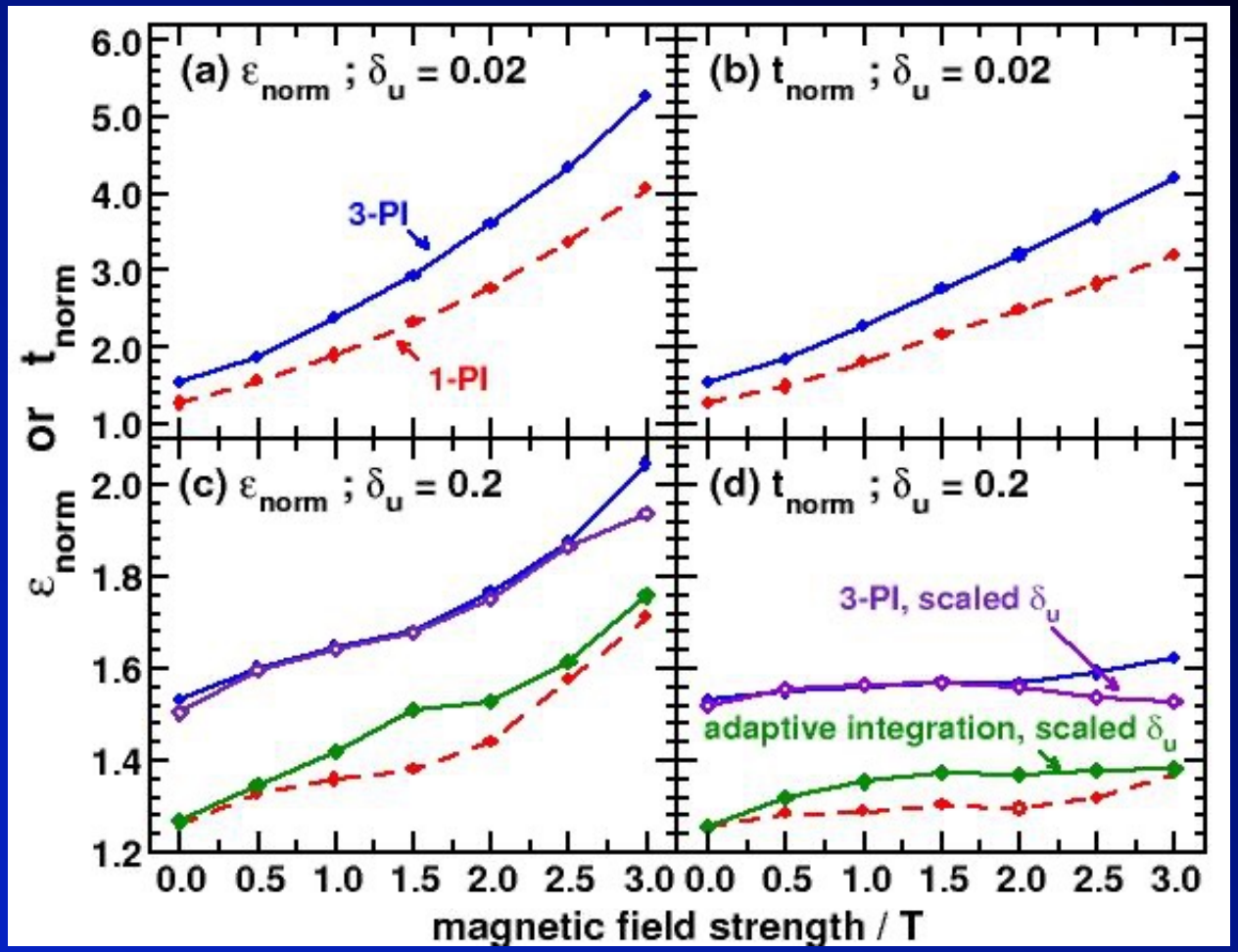
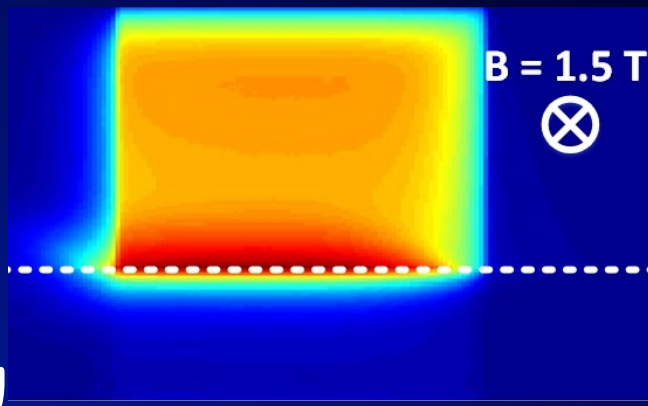


smaller t_{norm} better: ε_{norm} is inverted,

higher ε better => lower ε_{norm} is better

efficiency in phantom calculations

- $(30\ cm)^3$ phantom, $(3\ mm)^3$ voxels
- 6 MV beam with 1.5 T mag field
- 4x4 cm^2 field
- 2% stats on central axis
- on Intel Xeon 2680 2.5 GHz CPU
  - 30 min no mag field
  - 44 min 1.5 T mag field
- ion chamber calc for 2% stats
  - 6 s no mag field
  - 9 s with 1.5 T field



making it work with EGSnrc

- The EGSnrc manual (PIRS701) does not yet mention Malkov's macros, but the system appears to handle either set (EMF or EEMF).

- See section 3.14.1 in manual and use $(EGS_SOURCEDIR)EEMF_macros.mortran found on $HEN_HOUSE/src instead of $(EGS_SOURCEDIR)emf_macros.mortran when modifying SOURCES in any (except BEAMnrc) user_code.make file.

- For BEAMnrc, in sources.make use, $(EGS_SOURCEDIR)EEMF_macros_beamnrc.mortran

Similar changes for other apps as per PIRS701

define B field in .egsinp file

```
:start MC transport parameter:
        :
        :
# B-field defined in the X , Y, Z directions
Magnetic Field = 0, 1.5, 0        # magnetic field in tesla
EM ESTEPE = 0.2                   #default 0.02
        :
        :
:stop MC transport parameter:
```

Same for electric field in V/cm,
either standalone or combined

the end